# Golden-Ratio Step Size as a Conservative Heuristic for Discrete Optimization-Curve Convexity in Gradient Descent

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**Abstract**

We introduce a simple discrete diagnostic for the *optimization curve* of gradient descent, defined by the second forward difference of objective values along iterates. Motivated by a conservative engineering perspective, we propose the memorable step-size rule $\eta \approx \varphi/L$ (golden ratio over a Lipschitz gradient constant) and outline a reproducible evaluation protocol for empirical checks. **This note makes no universal theorem-level claim.** All theoretical discussion is explicitly heuristic or conjectural.

## 1 Scope and Non-Claims

This note is intentionally *not* a theorem paper. It is a compact, reproducible heuristic note.

**What is claimed.**

- A well-defined discrete diagnostic based solely on the observed sequence $g_k := f(x_k)$.
- A conservative step-size heuristic $\eta \approx \varphi/L$ with an optional safety factor.
- A concrete evaluation protocol that can be replicated across objectives and seeds.

**What is not claimed.**

- No statement that $\eta = \varphi/L$ guarantees discrete convexity of $\{g_k\}$ for all convex $L$-smooth $f$.
- No assertion that the golden ratio is optimal for any criterion.
- No implication that satisfying the diagnostic improves convergence rates or final performance.

## 2 Setup

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable and convex with $L$-Lipschitz gradient (i.e., $f$ is $L$-smooth). Consider gradient descent

$$x_{k+1} = x_k - \eta \nabla f(x_k), \qquad k = 0, 1, 2, \ldots \tag{1}$$

with constant step size $\eta > 0$. Define the *optimization curve*

$$g_k := f(x_k). \tag{2}$$

## 3 Discrete Optimization-Curve Convexity

**Definition 1** (Discrete second difference / convexity diagnostic)**.** For $k \geq 1$, define the discrete second difference of $\{g_k\}$ by

$$\Delta^2 g_k := g_{k-1} - 2g_k + g_{k+1}. \tag{3}$$

We say the optimization curve is *discretely convex* on an index set $\mathcal{K} \subseteq \{1, 2, \dots\}$ if $\Delta^2 g_k \geq 0$ for all $k \in \mathcal{K}$. In finite-precision experiments, we record a *violation* when

$$\Delta^2 g_k < -\varepsilon, \tag{4}$$

for a chosen tolerance $\varepsilon \geq 0$.

*Remark* 1. This is an *observational diagnostic* on $\{f(x_k)\}$. It is not equivalent to convexity of $f$, and does not by itself certify any convergence property.

## 4 A Conservative Step-Size Heuristic

Let $\varphi := \frac{1+\sqrt{5}}{2} \approx 1.618$ denote the golden ratio. We propose the step-size heuristic

$$\eta = \alpha \frac{\varphi}{L}, \qquad \alpha \in (0, 1], \tag{5}$$

where $\alpha$ is an optional safety factor (e.g., $\alpha \in [0.95, 0.99]$).

**Rationale (informal).** For $L$-smooth convex objectives, the classical descent stability window for fixed-step gradient descent is often presented as $\eta \in (0, 2/L)$ in standard analyses. The value $\varphi/L$ lies comfortably below $2/L$ and is easy to remember. The intent is a conservative default that empirically tends to yield "smooth" objective traces, as measured by the diagnostic in Definition 3–4.

**Conjecture 1** (Heuristic reliability (empirical))**.** *For many practical convex $L$-smooth objectives encountered in standard machine learning workflows, the choice $\eta \approx \varphi/L$ tends to exhibit fewer violations of* (4) *than more aggressive normalized step sizes such as* $1.7/L$–$1.8/L$*, under a fixed and disclosed evaluation protocol.*

## 5 Evaluation Protocol

To make empirical statements interpretable, any experiment should disclose at least:

**P1. Objective and preprocessing.** Exact definition of $f$, dataset, and normalization.

**P2. How $L$ is obtained.** Exact method (analytic, upper bound, backtracking proxy, spectral estimate, etc.).

**P3. Diagnostic tolerance.** The value of $\varepsilon$ in (4).

**P4. Initialization.** Distribution of $x_0$ and all random seeds.

**P5. Counting window.** Index set $\mathcal{K}$ where violations are counted (e.g., $k \in \{1, \dots, K\}$ or a burn-in excluded window).

**P6. Summary statistics.** Violation rate, distribution of $\min_{k \in \mathcal{K}} \Delta^2 g_k$, and optionally quantiles of $\Delta^2 g_k$.

## 6 Empirical Observations (Illustrative)

Under the above protocol, simple experiments on (i) quadratic objectives, (ii) least squares, and (iii) logistic regression (with standard normalization) can show that $\eta \approx \varphi/L$ is often associated with a low observed rate of violations, relative to step sizes closer to $2/L$ (e.g., $1.7/L$–$1.8/L$). These observations are context-dependent and should be interpreted as engineering evidence, not as a theorem.

## 7 Discussion

A theorem-level guarantee would require bounding the discrete second difference $\Delta^2 g_k$ using only convexity and $L$-smoothness while accounting for the two-step dependence induced by the recurrence (1). This note does not attempt such a proof and makes no claim of universality.

## 8 Practical Recommendation

When $L$ is known or reliably estimated and smooth optimization curves are desirable, a conservative default is

$$\eta = 0.99\,\frac{\varphi}{L}, \tag{6}$$

followed by empirical validation using the protocol in Section 5.

## Attribution

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